

Endogenous Education Choice in Segmented Search Markets

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Motivation

Structural search models of education and the labour market are typically estimated and analysed in steady state, or against business-cycle moments. Crises are not business cycles. The financial crisis and COVID shifted the structure of the labour market, not only its position in a cycle. A crisis is therefore a shift in structural parameters i.e. a different steady state. Furthermore, the financial crisis and COVID had different labour market signatures: Both events increased unemployment, but had opposite effects on vacancies. Two questions naturally follow: Which parameters shift during a crisis? How does the economy adjust between the resulting equilibria?

To answer those questions, I build an equilibrium model of two labour markets that workers search in separately, each with its own frictions and its own tightness, joined by a training decision. Workers are born with an aptitude for each market, drawn together and fixed for life, so who works where is settled by comparative advantage. Training develops no skill: it buys *access* to the market that rewards the aptitude a worker already has. The boundary between the two markets is therefore an equilibrium object, and where it sits depends on conditions in both markets relative to each other.

Research Question

What determines the training decision in a frictional, skill-segmented labour market, and how does it respond to aggregate shocks over time?

I estimate the model on a pair of time windows before and after each crisis. Within a pair I hold fixed the parameters that describe the population i.e. the joint distribution of the two aptitudes, as well as the parameters that describe workers' outside options, on the ground that neither the ability composition of prime-age workers nor unemployment income have changed during either of the crises. Everything else is re-estimated before and after the event, and the re-estimated parameters pin down which channels moved during the event and by how much. Given the two estimated equilibria, model-implied transition paths between them are then drawn, tracing the adjustment over time.

Model

The environment is continuous-time and builds on the search-and-matching tradition (Diamond, 1982; Mortensen and Pissarides, 1994; Pissarides, 2000), with two segmented markets, unskilled

and skilled, each with its own matching function, its own free-entry condition and its own tightness. Four features distinguish the setup:

- **Two aptitudes, and comparative advantage.** A worker is born with an aptitude for each market, the pair drawn jointly from a distribution with a Gaussian copula and fixed for life. Production is linear and strictly increasing in own aptitude in both markets, so who works where is settled by comparative advantage rather than by curvature in any technology. The dependence between the two aptitudes is the primitive that governs how workers divide between the markets, as in (Gola, 2021), but with search frictions on both sides of the boundary.
- **Education is market access, not skill.** Training develops no productive capacity. It buys entry to the market that rewards the aptitude a worker already holds. The boundary between the two markets is therefore an equilibrium object, and where it sits depends on conditions in both markets relative to each other. This is the paper’s departure from search models of education choice, in which education raises what a worker is worth (Charlot and Decreuse, 2005, 2010; Charlot and Malherbet, 2013; Bárány, 2016).
- **The boundary moves in both directions.** A worker crosses into the skilled market through training; she also crosses back out when her comparative advantage lies on the unskilled side. Crossing back abandons the training, and she returns to the unskilled market as though she had never trained.
- **Different frictions on the two sides.** In the unskilled market a match begins at the top of the quality support and each shock replaces quality with an independent draw. In the skilled market match quality is drawn on contact, workers search on the job, and the ladder they climb is one in match quality, following Pissarides (2000). Wages are bargained in both markets.

The general-equilibrium feedback: A shock to one market changes its tightness, which changes the continuation value of searching there, which moves the boundary, which changes the composition of both search pools, which changes both tightnesses through free entry. Nothing in that feedback loop can be solved in one market at a time.

Data and Estimation Strategy

The model is estimated by simulated method of moments on U.S. data for workers aged 16–64, with the skilled market defined by attainment of a bachelor’s degree or above. Moments come from the CPS basic monthly and the CPS Annual Social and Economic Supplement, with job-to-job flows from Census data, vacancies from JOLTS, wage-change hazards from the SIPP, and enrolment from the NCES. The moments targeted: unemployment and tightness in each market, the wage distribution, the composition of employment, the job-finding, separation and job-to-job rates, and the training share.

A crisis is estimated as a *pair* of stationary equilibria. Each event contributes a pre-event baseline window and a post-event window:

Window	Time Period
FC baseline	2003–2007
FC crisis	2008–2010
COVID baseline	2015–2019
COVID crisis	2020–2022

Within a pair, the parameters describing the population — the joint distribution of the two aptitudes — and the parameters describing workers' unemployment income are held fixed, on the ground that neither the aptitude composition of the workforce nor unemployment income changed over the span of either event. Everything else is re-estimated on the crisis window. The re-estimated parameters are what identifies which channels moved and by how much.

Results

The model reproduces the unemployment rates, the tightnesses, the wage distributions and the skill composition of the economy it is estimated on, in both events, with no productivity gain from training. The mechanism is comparative advantage and market access alone.

The two events separate on the estimated parameters in the way their labour-market signatures suggest. The financial-crisis pair moves through the demand channels — matching efficiency and the cost of posting a vacancy — which traces a movement along the Beveridge curve. The COVID pair additionally moves the skilled market's separation margin, which shifts the curve outwards.

Contributions

1. **A model of two frictional markets joined by an education decision.** The return to training is a difference between two continuation values, each governed by how long a worker waits for an offer in that market, what the offer is worth, and how long it lasts. The boundary is open in both directions: a worker crosses in by paying for access, and crosses back out when her comparative advantage lies on the unskilled side. Crossing back out abandons the training: she returns to the unskilled market as though she had never trained.
2. **Estimation on a pair of equilibria, with the partition argued.** The parameters describing the population's innate ability and unemployment income are held fixed over the span of a pair. The innate talents of people cannot change overnight, and unemployment income is institutional: unemployment benefits and universal basic income are government policies, and neither changed during either crisis. Holding those fixed buys precision on the parameters that are estimated on the post-event time windows.
3. **Transition between two estimated equilibria.** With an estimated equilibrium at each end of the event, the path the economy takes from one to the other can be traced.

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